

Ad Hoc Human–AI Orchestration via Skill Inference

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Infer the latent skills of humans and AI from a few noisy observations well enough for orchestration

Introduction

Orchestrating ad hoc human–AI teams means knowing who is best at what. But résumés and model cards don't compare across people and AI, and testing everyone is slow, costly, and soon stale. We need one set of skill axes for both, filled from a few observations and kept current. Skill inference learns these skill profiles efficiently from how skills correlate.

Method

Represent each entity, human or AI, as a Gaussian belief $\mathcal{N}(\mu, \Sigma)$ over shared skills. From the population average, observe one noisy rating y_j ; a Kalman update spreads it to every skill through Σ :

μ : skill profile

Σ : skill covariance (uncertainty)

$$\begin{aligned} \mathbf{K} &= \Sigma_{:,j} / (\Sigma_{jj} + \sigma_{\text{obs}}^2) && \text{Kalman gain} \\ \mu &\leftarrow \mu + \mathbf{K}(y_j - \mu_j) && \text{spread to all skills} \\ \Sigma &\leftarrow \Sigma - \mathbf{K}\Sigma_{j,:} && \text{shrink uncertainty} \end{aligned}$$

Conclusion

- One Bayesian method works across humans, LLMs, and mixed pools.
- Because skills correlate, each observation updates the whole skill profile through Σ , recovering faster than independent estimation.
- The posteriors stay calibrated at $\approx 95\%$ under an entity-level prior.
- Online routing on the posteriors finds human + AI complementarity and scales to large teams.

Future work

- Deploy in production on human–AI workflows.
- Extend the profiles beyond skills with preferences, beliefs, and goals.

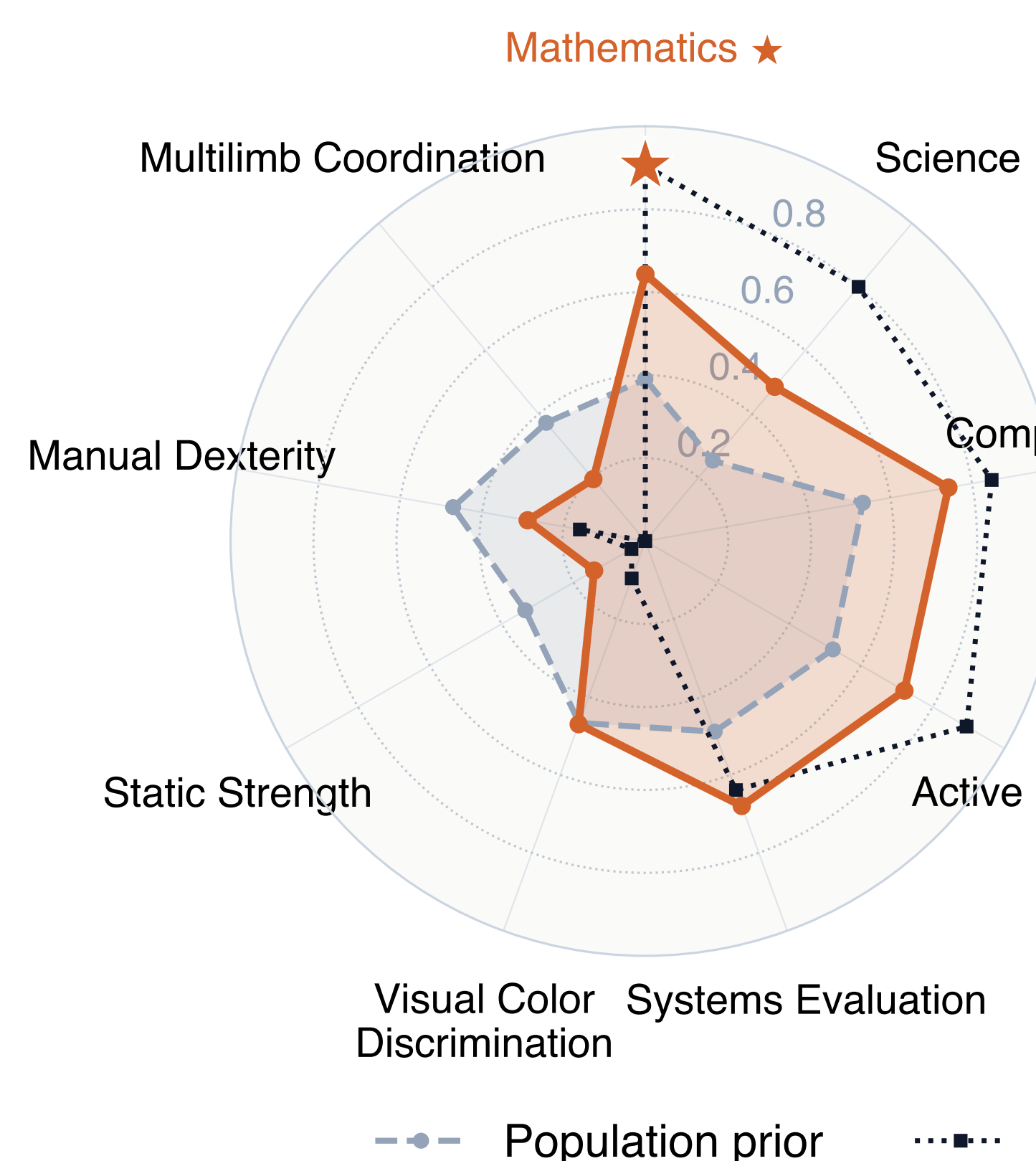
References

- Masters et al. Orchestrating human-AI teams: the manager agent as a unifying research challenge, 2025.
- Prunty et al. Reverse Turing tests for human-machine task suitability should be profile-driven, 2026.
- Zhou et al. General scales unlock AI evaluation with explanatory and predictive power. Nature, 2026.

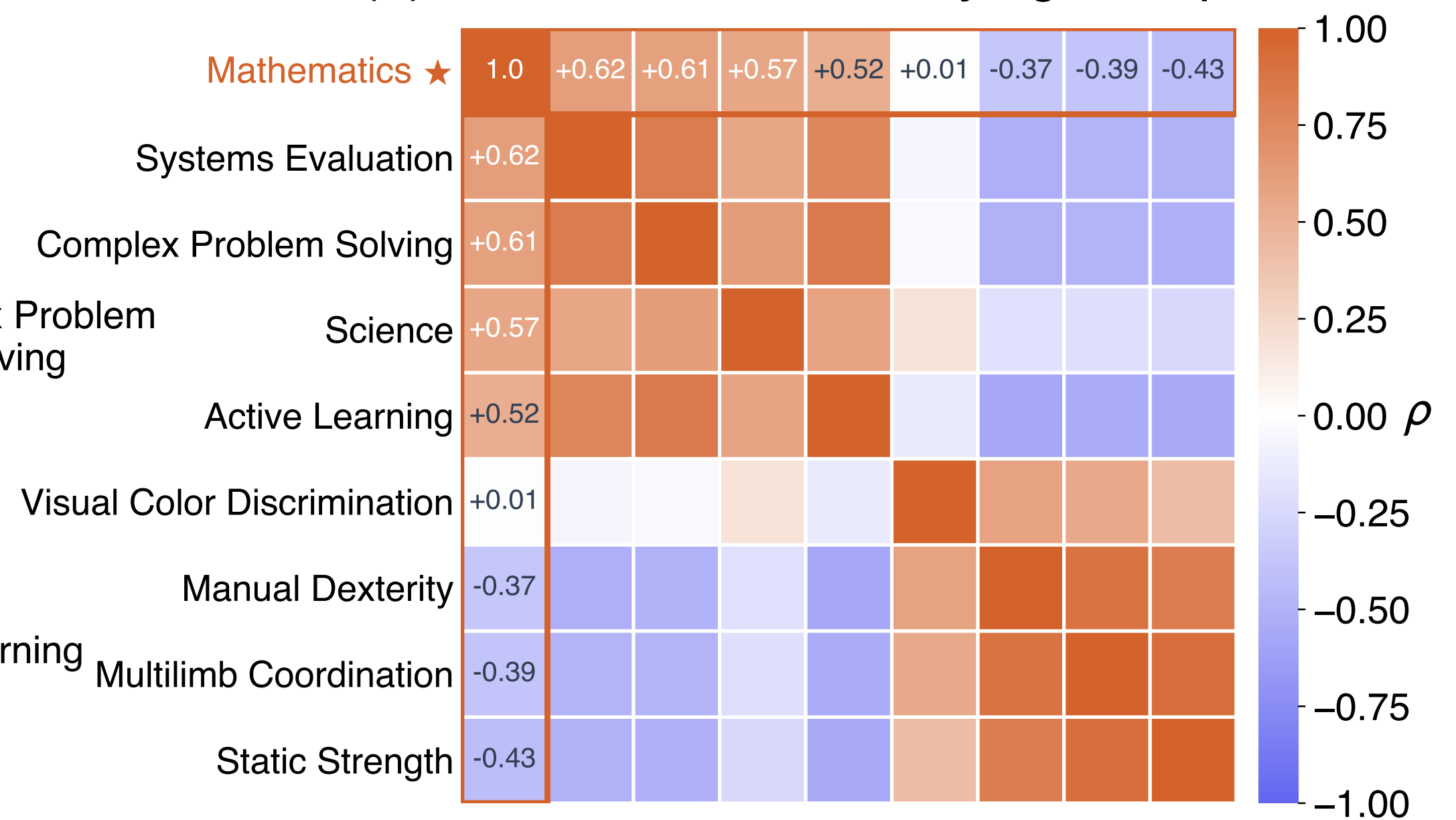
How it works

A A WORKED UPDATE VIA THE COVARIANCE

(a) Skill profile: prior vs. posterior



(b) Skill correlations carrying the update

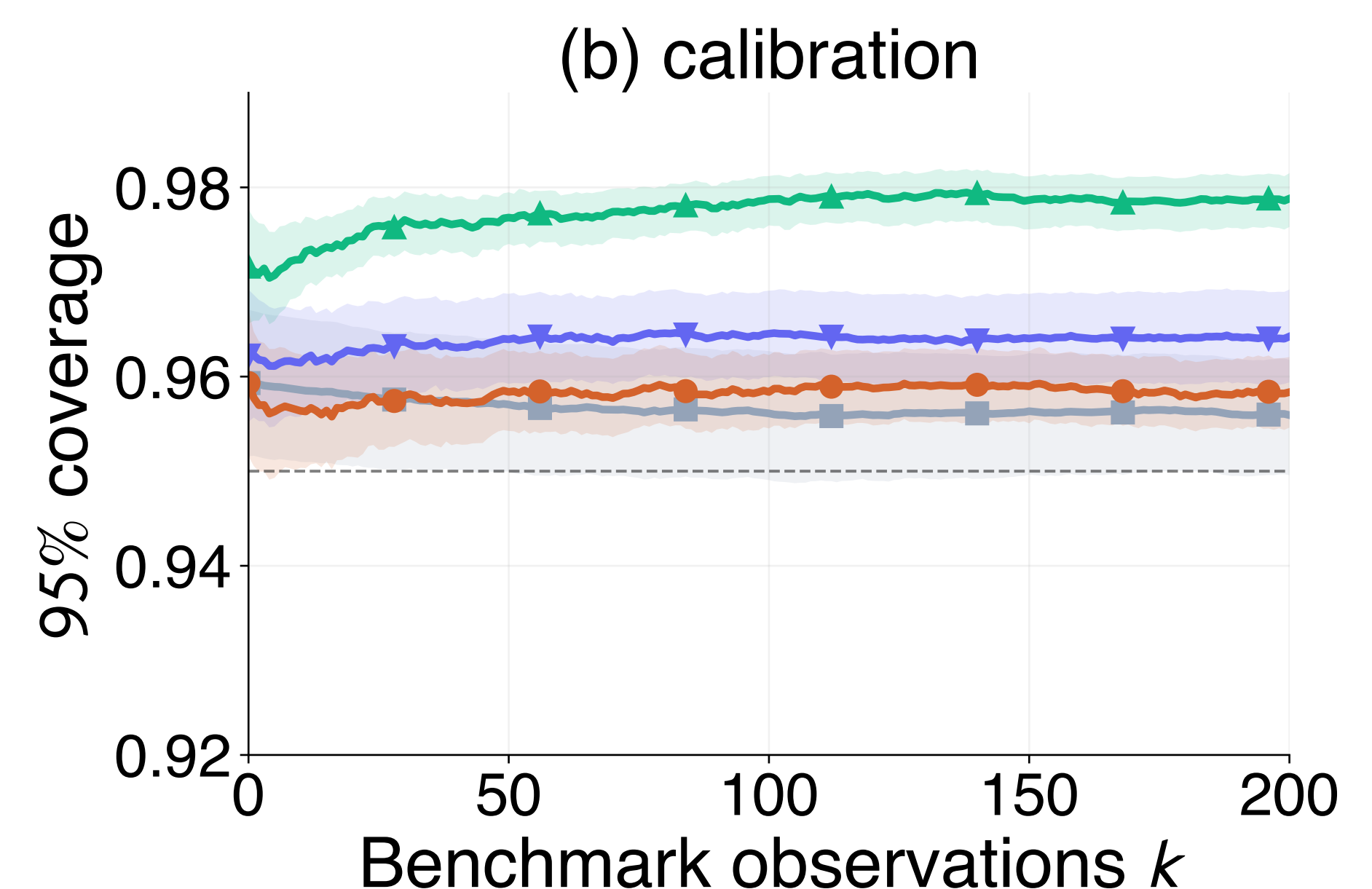
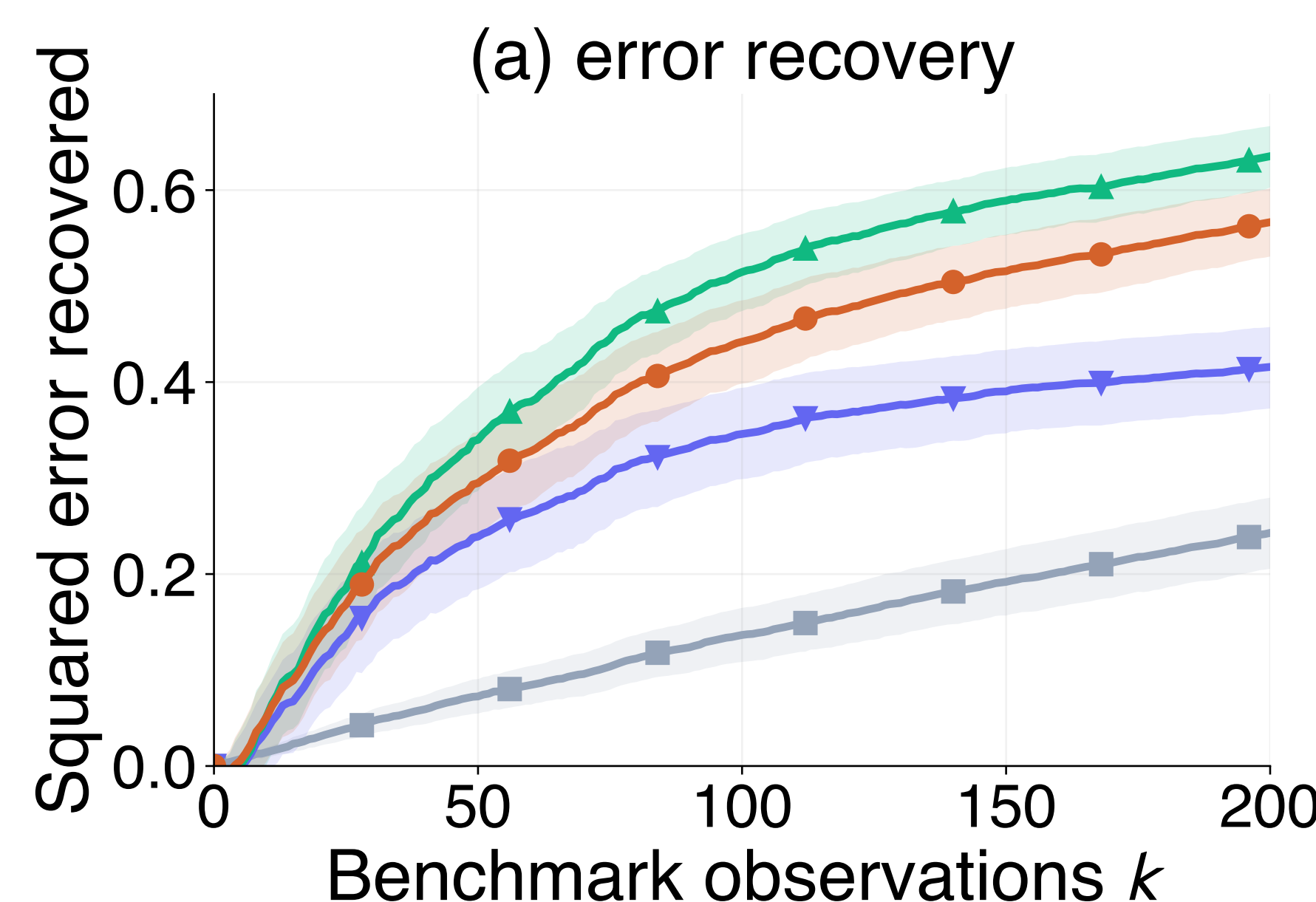


Population prior (dashed blue line), Target (true) (dotted black line), Posterior (solid orange line), Observation $y=0.90$ (star)

One observation ($y=0.90$) updates the whole skill profile via Σ .

Results

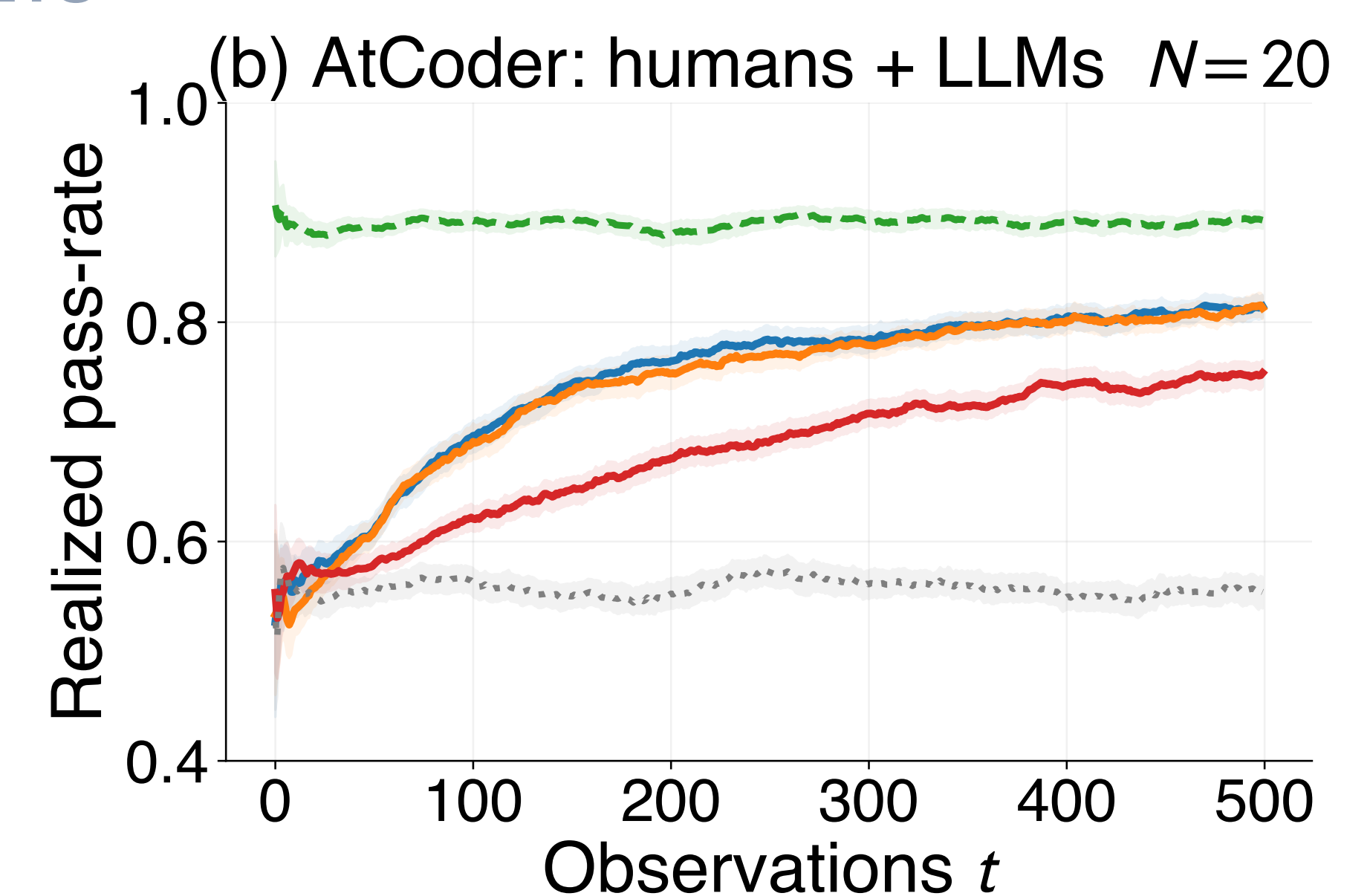
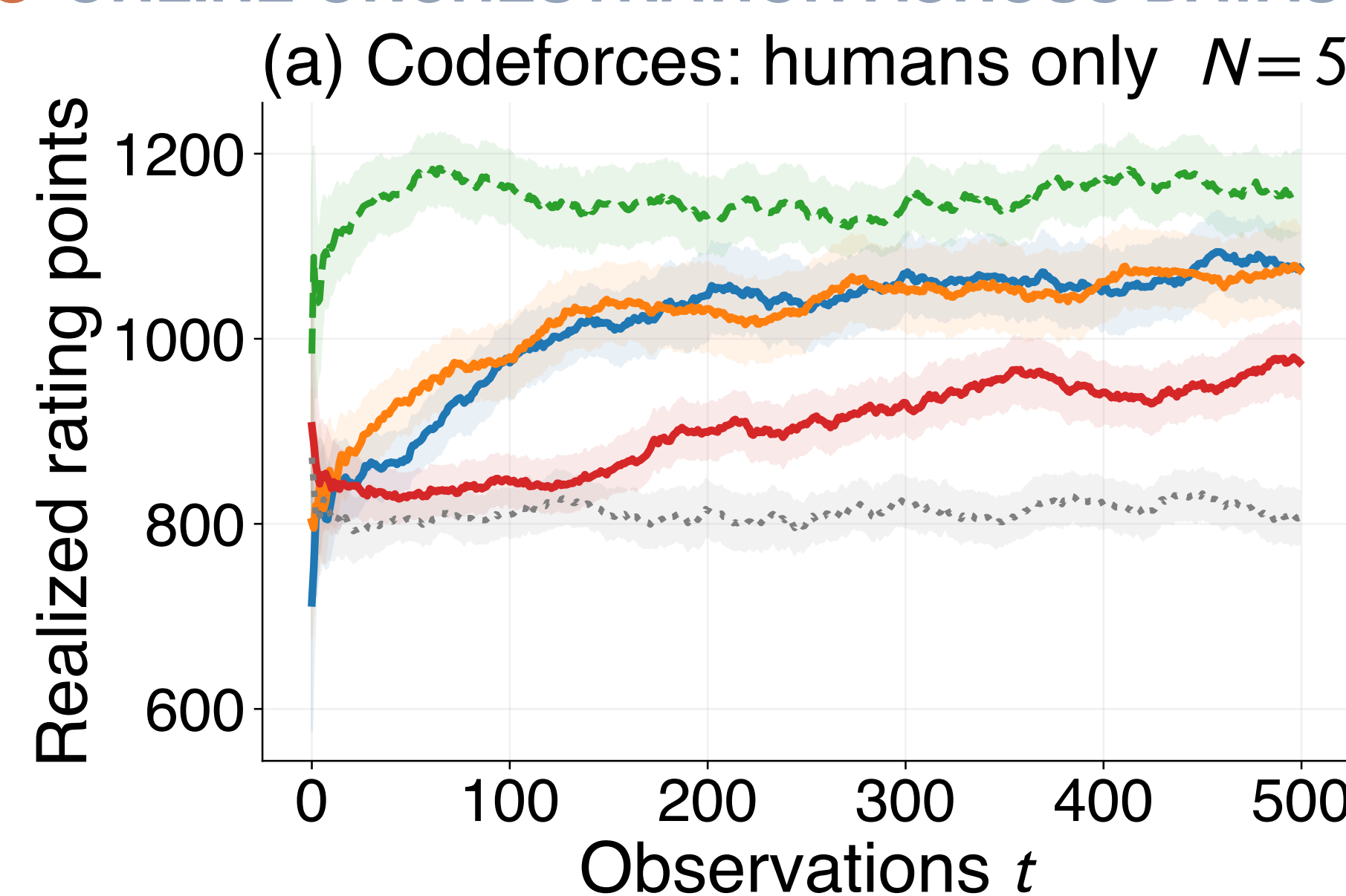
B RECOVERY & CALIBRATION ON THE OPEN LLM LEADERBOARD



Diagonal (baseline) (grey line), IRT-1D (baseline) (blue line), GMM- Σ (ours) (green line), Σ full (ours) (orange line)

Recovers skill profiles faster than Diagonal and IRT; calibrated at $\approx 95\%$.

C ONLINE ORCHESTRATION ACROSS DATASETS

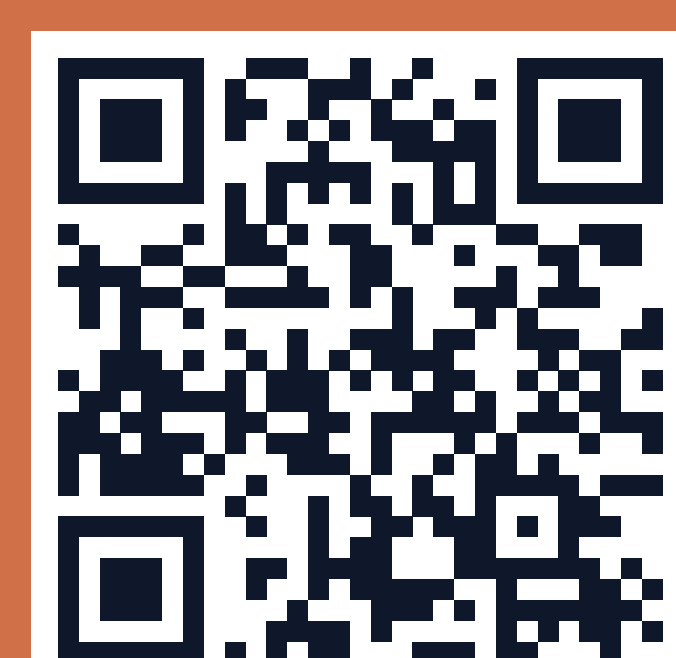


Σ (ours) (blue line), GMM- Σ (ours) (orange line), diagonal- Σ (baseline) (red line), Oracle (dashed green line), Random (dotted grey line)

Routing on the posteriors beats independent estimation on both datasets.

Open source

`pip install skillinfer` • github.com/kostadindev/skillinfer
kostadindev.github.io/skillinfer • tailorworkflow.com • kgd26@cam.ac.uk



skillinfer
library & docs